A2. Boundary Value Analysis

Boundary Value Analysis (BVA) focuses on values at the edges of an input range because defects commonly occur at these points.

We will test:

- Current credits: 0–18.
- Course credits: 1–4.
- Combined credits: maximum 18.
- Course enrollment: 0–capacity.
- Outstanding dues: minimum 0.

BVA test cases

Test ID	Requirement	Input	Expected output
TC20	FR-02	Current credits = -1	ValueError
TC21	FR-01	Current credits = 0, course = 1	True
TC22	FR-01	Current credits = 18, course = 1	False
TC23	FR-02	Current credits = 19	ValueError
TC24	FR-02	Course credits = 0	ValueError
TC25	FR-01	Course credits = 1	Valid input

TC26	FR-01	Course credits = 4	Valid input
TC27	FR-02	Course credits = 5	ValueError
TC28	FR-01	Current = 15, course = 3	True
TC29	FR-01	Current = 16, course = 3	False
TC30	FR-08	Capacity = 30, enrolled = 29	REGISTERED
TC31	FR-08	Capacity = 30, enrolled = 30	WAITLISTED
TC32	FR-15	Capacity = 30, enrolled = 31	ValueError
TC33	FR-15	Capacity = 0	ValueError
TC34	FR-04	Outstanding dues = 0	Registration allowed if otherwise eligible
TC35	FR-04	Outstanding dues = 1	DUES_BLOCK

Important boundary observations

1. Combined credit limit

- 15 + 3 = 18 → True
- 16 + 3 = 19 → False

Exactly 18 credits are allowed according to FR-01.

2. Course capacity

- 29/30 → Seat available.
- 30/30 → Course full, student should be waitlisted.
- 31/30 → Invalid enrollment.

3. Outstanding dues

- Rs. 0 → Registration can proceed.
- Rs. 1 → Registration must be blocked.